



**INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS
ACADEMY (ICDFA)**

SBT-DF202 · PRACTICAL LABORATORIES · LAB 3
DATA CARVING WITH XXD, BINWALK AND SCALPEL

STUDENT NAME: IBRAHIM ISHAKU
STUDENT ID: 2025/FWSD/11334

**FELLOWSHIP IN WEB APPLICATION SECURITY & DIGITAL
FORENSICS**

COURSE: BASIC COMPUTER SKILLS FOR DIGITAL FORENSICS
DATA CARVING WITH XXD, BINWALK AND SCALPEL

INSTRUCTOR: AMINU IDRIS, AMCPN

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Acknowledgement

I thank my instructor for the comprehensive Data Carving and File Recovery Techniques materials that formed the foundation for this lab. The practical walkthrough on xxd, Binwalk, Scalpel, file signatures, and hexadecimal analysis gave me the essential skills required for this practical exercise. I also appreciate the International Cybersecurity and Digital Forensics Academy (ICDFA) for providing the forensic images and lab environment for this practical assessment.



Certification / Declaration

I, **Ibrahim Ishaku**, with Student ID: **2025/FWSD/11334**, hereby declare that this lab report titled "**Data Carving with XXD, Binwalk and Scalpel**" is my original work. All forensic analysis, hexadecimal inspection, data carving, and hash verification tasks documented in this report were performed by me in the ICDFA Kali Linux lab environment. The answers are based on my actual practical work and understanding of the course materials. I confirm that the original evidence files were not modified during the investigation and that all recovered evidence, screenshots, and command outputs were stored separately from the source files.

Student

Signature: _____

Date: 4th September, 2026



Instructor Approval Page

Instructor Name:

Comments / Evaluation:

Instructor Signature: _____

Date: _____



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Executive Summary

This lab report documents my practical work on Data Carving with XXD, Binwalk and Scalpel as required for the SBT-DF202 Practical Laboratories course. I completed a comprehensive forensic data carving exercise using the provided evidence files. I performed hexadecimal analysis using xxd to identify JPEG file signatures (SOI and EOI), reconstructed an image from a hex dump, and verified the reconstruction using cryptographic hashes. I examined embedded metadata-like content using xxd and strings. I used The Sleuth Kit tools (img_stat, mmls, fsstat, fls, icat, blkcat) to examine data units. I used binwalk to identify embedded file structures. I extracted and prepared the USB image from 120M.7z and ran scalpel to carve deleted files. All recovered files were verified using cryptographic hashes to maintain evidence integrity. Through this lab, I gained practical skills in data carving, file signature identification, and forensic recovery techniques essential for digital forensics professionals.



1.0 Section 1 – Introduction

Data carving is a forensic technique used to recover files and data from storage media when file system metadata is missing, damaged, or unreliable. Unlike normal file recovery, which relies on file system structures, data carving identifies files by their **signatures** – specific byte patterns that appear at the beginning (headers) and end (footers) of files.

This lab covers the use of three key tools:

- i. **xxd** – Creates hexadecimal dumps and reverse dumps for manual file reconstruction.
- ii. **Binwalk** – Identifies embedded files and signatures within binary data.
- iii. **Scalpel** – Carves files from raw data using a configuration file.

The lab uses three evidence files:

- i. J_ub_law.jpg – JPEG image for hex and signature analysis.
- ii. Ch01InChap01.dd – Forensic DD image for inode analysis.
- iii. 120M.7z – Compressed USB image for Scalpel file carving.

2.0 Section 2 – Preparation and Evidence Handling

2.1 Create Forensic Working Directory

```
mkdir -p ~/SBT-DF202_Lab3_{evidence,recovered,reports,screenshots,Logs}
cd ~/SBT-DF202_Lab3_evidence
pwd
ls -la
```



```
ibrahim@kali: ~/SBT-DF202_Lab3_evidence
Session Actions Edit View Help
Documents SBT-DF202_Lab2_Images
Downloads Templates
logs Videos
ch00inChap01
ch00inChap01
ch00inChap01

(ibrahim@kali)-[~]
$ mkdir -p ~/SBT-DF202_Lab3_{evidence,recovered,reports,screenshots,logs}
cd ~/SBT-DF202_Lab3_evidence
pwd
ls -la

/home/ibrahim/SBT-DF202_Lab3_evidence
total 8
drwxrwxr-x  2 ibrahim ibrahim 4096 Sep  4 02:49 .
drwx----- 37 ibrahim ibrahim 4096 Sep  4 02:49 ..

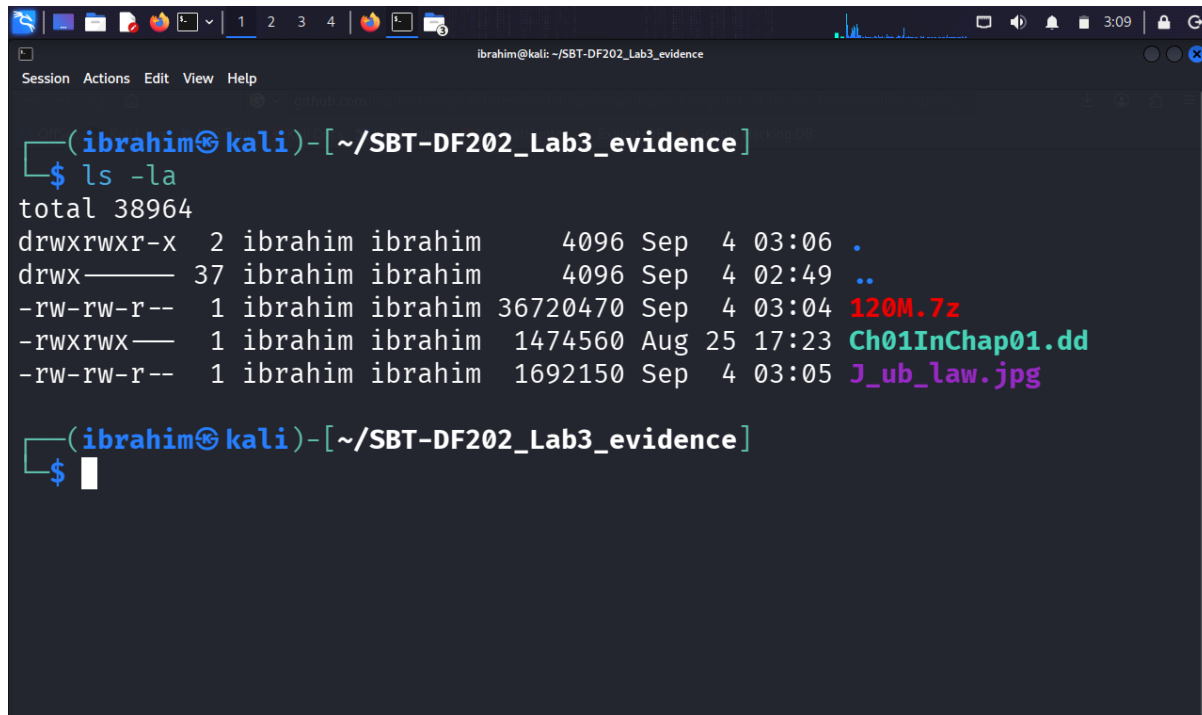
(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$
```

Figure 1.1: Lab folder structure created successfully.

2.2 Download Evidence Files

Required Files:

- i. *J_ub_law.jpg* – JPEG Sample
- ii. *Ch01InChap01.dd* – Forensic DD Image
- iii. *120M.7z* – USB Image for Scalpel



```

(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$ ls -la
total 38964
drwxrwxr-x  2 ibrahim ibrahim   4096 Sep  4 03:06 .
drwx----- 37 ibrahim ibrahim   4096 Sep  4 02:49 ..
-rw-rw-r--  1 ibrahim ibrahim 36720470 Sep  4 03:04 120M.7z
-rwxrwx---  1 ibrahim ibrahim 1474560 Aug 25 17:23 Ch01InChap01.dd
-rw-rw-r--  1 ibrahim ibrahim 1692150 Sep  4 03:05 J_ub_law.jpg
  
```

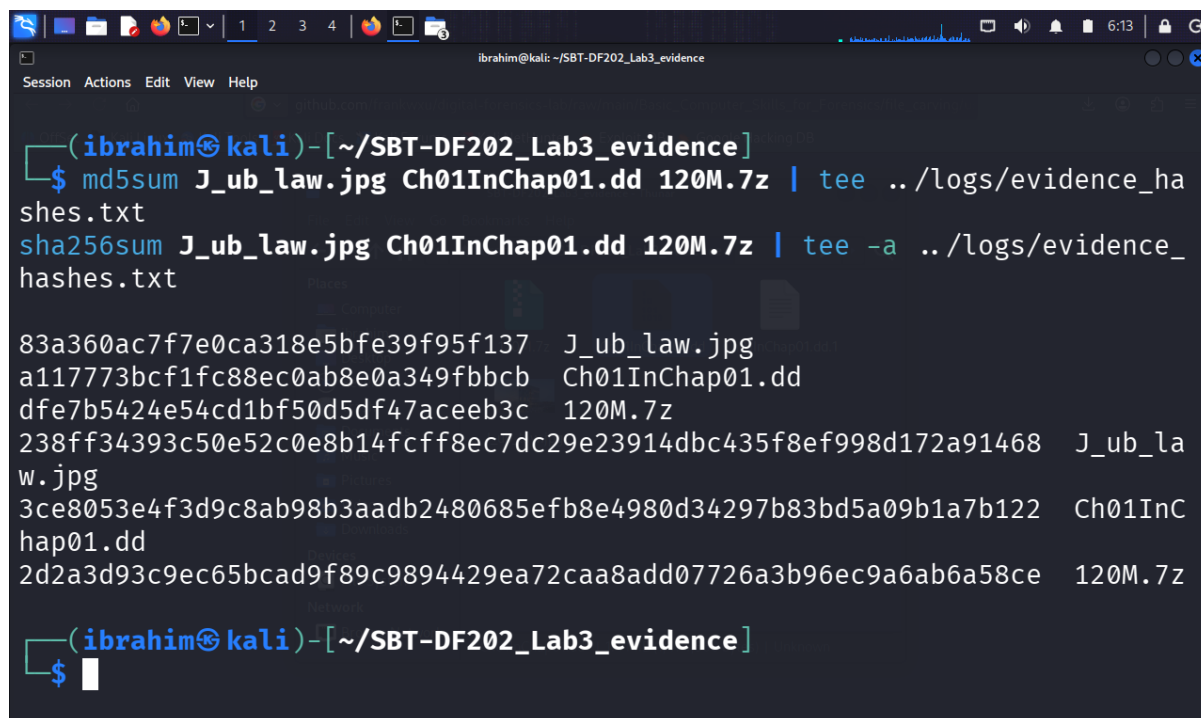
Figure 1.2: Evidence files downloaded.

2.3 Record Evidence Information

Attribute	Value
Evidence File 1	J_ub_law.jpg
Evidence File 2	Ch01InChap01.dd
Evidence File 3	120M.7z
Storage Location	~/SBT-DF202_Lab3_evidence/

2.4 Calculate Evidence Hashes

```
md5sum J_ub_Law.jpg Ch01InChap01.dd 120M.7z | tee ../logs/evidence_hashes.txt
sha256sum J_ub_Law.jpg Ch01InChap01.dd 120M.7z | tee -a ../logs/evidence_hashes.txt
```



```
(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$ md5sum J_ub_Law.jpg Ch01InChap01.dd 120M.7z | tee ../logs/evidence_hashes.txt
sha256sum J_ub_Law.jpg Ch01InChap01.dd 120M.7z | tee -a ../logs/evidence_hashes.txt

83a360ac7f7e0ca318e5bfe39f95f137 J_ub_Law.jpg
a117773bcf1fc88ec0ab8e0a349fbbcb Ch01InChap01.dd
dfe7b5424e54cd1bf50d5df47aceeb3c 120M.7z
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468 J_ub_Law.jpg
3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122 Ch01InChap01.dd
2d2a3d93c9ec65bcad9f89c9894429ea72caa8add07726a3b96ec9a6ab6a58ce 120M.7z

(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$
```

Figure 1.3: MD5 and SHA-256 hashes of the evidence files.

File	MD5 Hash	SHA-256 Hash
J_ub_Law.jpg	83a360ac7f7e0ca318e5bfe39f95f137	238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468
Ch01InChap01.dd	a117773bcf1fc88ec0ab8e0a349fbbcb	3ce8053e4f3d9c8ab98b3aadb2480685efb8e4980d34297b83bd5a09b1a7b122
120M.7z	dfe7b5424e54cd1bf50d5df47aceeb3c	2d2a3d93c9ec65bcad9f89c9894429ea72caa8add07726a3b96ec9a6ab6a58ce

3.0 Section 3 – Part A: File Signatures and Hex Analysis

3.1 JPEG File Signatures

Signature	Hex Value	Description
SOI (Start of Image)	FF D8	JPEG header signature
EOI (End of Image)	FF D9	JPEG footer signature

3.2 Inspect J_ub_law.jpg with xxd

`xxd J_ub_law.jpg | head -20`

```

(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$ xxd J_ub_law.jpg | head -20
00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00  ....Exif..II*.
00000010: 0800 0000 1000 0001 0300 0100 0000 4013  ....@.
00000020: 0000 0101 0300 0100 0000 d00c 0000 0201  ....
00000030: 0300 0300 0000 ce00 0000 0601 0300 0100  ....
00000040: 0000 0200 0000 0f01 0200 1200 0000 d400  ....
00000050: 0000 1001 0200 0900 0000 e600 0000 1201  ....
00000060: 0300 0100 0000 0100 0000 1501 0300 0100  ....
00000070: 0000 0300 0000 1a01 0500 0100 0000 ef00  ....
00000080: 0000 1b01 0500 0100 0000 f700 0000 2801  ....(
00000090: 0300 0100 0000 0200 0000 3101 0200 2100  ....1...!.
000000a0: 0000 ff00 0000 3201 0200 1400 0000 2001  ....2.....
000000b0: 0000 9882 0200 0c00 0000 3401 0000 9e9c  ....4.....
000000c0: 0100 5200 0000 4001 0000 6987 0400 0100  ..R...@...i....
000000d0: 0000 9401 0000 4404 0000 0800 0800 0800  ....D.....
000000e0: 4e49 4b4f 4e20 434f 5250 4f52 4154 494f  NIKON CORPORATIO

```

```

ibrahim@kali: ~/SBT-DF202_Lab3_evidence
Session Actions Edit View Help
00000050: 0000 1001 0200 0900 0000 e600 0000 1201 .....
00000060: 0300 0100 0000 0100 0000 1501 0300 0100 .....
00000070: 0000 0300 0000 1a01 0500 0100 0000 ef00 .....
00000080: 0000 1b01 0500 0100 0000 f700 0000 2801 .....(
00000090: 0300 0100 0000 0200 0000 3101 0200 2100 .....1...!.
000000a0: 0000 ff00 0000 3201 0200 1400 0000 2001 .....2.....
000000b0: 0000 9882 0200 0c00 0000 3401 0000 9e9c .....4.....
000000c0: 0100 5200 0000 4001 0000 6987 0400 0100 ..R...@...i....
000000d0: 0000 9401 0000 4404 0000 0800 0800 0800 .....D.....
000000e0: 4e49 4b4f 4e20 434f 5250 4f52 4154 494f NIKON CORPORATIO
000000f0: 4e00 4e49 4b4f 4e20 4434 00c0 c62d 0010 N.NIKON D4...-..
00000100: 2700 00c0 c62d 0010 2700 0041 646f 6265 '....-..'..Adobe
00000110: 2050 686f 746f 7368 6f70 2032 312e 3120 Photoshop 21.1
00000120: 284d 6163 696e 746f 7368 2900 3230 3230 (Macintosh).2020
00000130: 3a30 383a 3230 2031 303a 3230 3a30 3500 :08:20 10:20:05.

(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$

```

Figure 2.1: JPEG header (SOI) showing FF D8.

`xxd J_ub_law.jpg | tail -20`

```

ibrahim@kali: ~/SBT-DF202_Lab3_evidence
Session Actions Edit View Help

(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$ xxd J_ub_law.jpg | tail -20
0019d0c0: 2391 72f6 045b d7a7 d2d6 00fb 4f29 ff00 #.r..[.....0)..
0019d0d0: 194f cffc 1d09 7664 57db ae9f cd74 d3f9 .O....vdW....t..
0019d0e0: ff00 87aa d8cb 53bd 2bb4 88cd e3b3 93a6 .....S.+.....
0019d0f0: 68a3 9ea3 5484 ac48 7d4c 7493 e91f e361 h...T..H}Lt....a
0019d100: c5fd ac15 e271 d069 f0dd 324f 4528 3e29 .....q.i..20E(>)
0019d110: c78a c8a1 8ca1 5623 2e93 a63f 347c 298e .....V#...?4|).
0019d120: d6be 9b0e 4137 fa68 f4de aeb2 d3f9 ab54 ....A7.h.....T
0019d130: ac92 c2f3 6858 68a9 669b 4ad5 12ea b314 ....hXh.f.J.....
0019d140: 9471 c051 235f d2c3 817b 7bf6 47ce bd5a .q.Q#_...{.G..Z
0019d150: bdbf 6752 0186 378d 7cb3 06f2 9f5c 28ea ..gR..7.|....\(.
0019d160: ecfe 3d52 1918 3008 da88 b489 7f22 feae ..=R..0....."..
0019d170: 2def 629c 7ab5 16bc 7acd 5750 6ba9 63a7 -.b.z...z.WPk.c.
0019d180: 9155 0524 f1a8 648f 4d6a f9c7 95e4 8981 .U.$..d.Mj.....
0019d190: 55d6 1519 8293 6360 41e0 fbd5 0530 3aa9 U.....c`A....0:.
0019d1a0: 0bab 554f 4d91 346a 25a7 99d9 2a52 a1bd ..UOM.4j%...*R..

```

```

Session Actions Edit View Help
0019d110: c78a c8a1 8ca1 5623 2e93 a63f 347c 298e .....V# ... ?4|).
0019d120: d6be 9b0e 4137 fa68 f4de aeb2 d3f9 ab54 ....A7.h.....T
0019d130: ac92 c2f3 6858 68a9 669b 4ad5 12ea b314 ....hXh.f.J.....
0019d140: 9471 c051 235f d2c3 817b 7bf6 47ce bd5a .q.Q#_ ... {{.G..Z
0019d150: bdbf 6752 0186 378d 7cb3 06f2 9f5c 28ea ..gR..7.|....\(.
0019d160: ecfe 3d52 1918 3008 da88 b489 7f22 feae ..=R..0....." ..
0019d170: 2def 629c 7ab5 16bc 7acd 5750 6ba9 63a7 -.b.z ... z.WPk.c.
0019d180: 9155 0524 f1a8 648f 4d6a f9c7 95e4 8981 .U.$..d.Mj.....
0019d190: 55d6 1519 8293 6360 41e0 fbd5 0530 3aa9 U.....c`A....0:.
0019d1a0: 0bab 554f 4d91 346a 25a7 99d9 2a52 a1bd ..UOM.4j%...*R..
0019d1b0: 294f ae4a a859 c9b4 2885 515a f676 3a80 )O.J.Y..(.QZ.v:.
0019d1c0: fc7f 81df 1c79 f5e1 dc0f 53e3 a4d5 5f45 .....y....S..._E
0019d1d0: 182a d52d 2406 abc1 ad69 9479 0789 d1e1 .*.-$....i.y....
0019d1e0: ff00 5254 0938 040e 6ff5 f75a 9a54 f565 ..RT.8..o..Z.T.e
0019d1f0: d351 9ebf ffd9 .Q....

(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$

```

Figure 2.2: JPEG footer (EOI) showing FF D9.

Key Findings:

- i. Header signature: FF D8 (JPEG SOI)
- ii. Footer signature: FF D9 (JPEG EOI)

3.3 Create a Hexadecimal Dump

```

xxd J_ub_Law.jpg > J_ub_Law.hex
head -10 J_ub_Law.hex

```

```

(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$ xxd J_ub_law.jpg > J_ub_law.hex
head -10 J_ub_law.hex

00000000: ffd8 ffe1 17f0 4578 6966 0000 4949 2a00  ....Exif..II*.
00000010: 0800 0000 1000 0001 0300 0100 0000 4013  ....@.
00000020: 0000 0101 0300 0100 0000 d00c 0000 0201  ....
00000030: 0300 0300 0000 ce00 0000 0601 0300 0100  ....
00000040: 0000 0200 0000 0f01 0200 1200 0000 d400  ....
00000050: 0000 1001 0200 0900 0000 e600 0000 1201  ....
00000060: 0300 0100 0000 0100 0000 1501 0300 0100  ....
00000070: 0000 0300 0000 1a01 0500 0100 0000 ef00  ....
00000080: 0000 1b01 0500 0100 0000 f700 0000 2801  ....(
00000090: 0300 0100 0000 0200 0000 3101 0200 2100  ....1...!.

(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$

```

Figure 2.3: Plain hexadecimal dump of the image.

3.4 Reconstruct the Image from Hex Dump

```

xxd -r J_ub_law.hex J_ub_law_reconstructed.jpg
ls -lh J_ub_law*.jpg

```

```

(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$ xxd -r J_ub_law.hex J_ub_law_reconstructed.jpg
ls -lh J_ub_law*.jpg

-rw-rw-r-- 1 ibrahim ibrahim 1.7M Sep  4 03:05 J_ub_law.jpg
-rw-rw-r-- 1 ibrahim ibrahim 1.7M Sep  4 19:12 J_ub_law_reconstructed.jpg

(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$

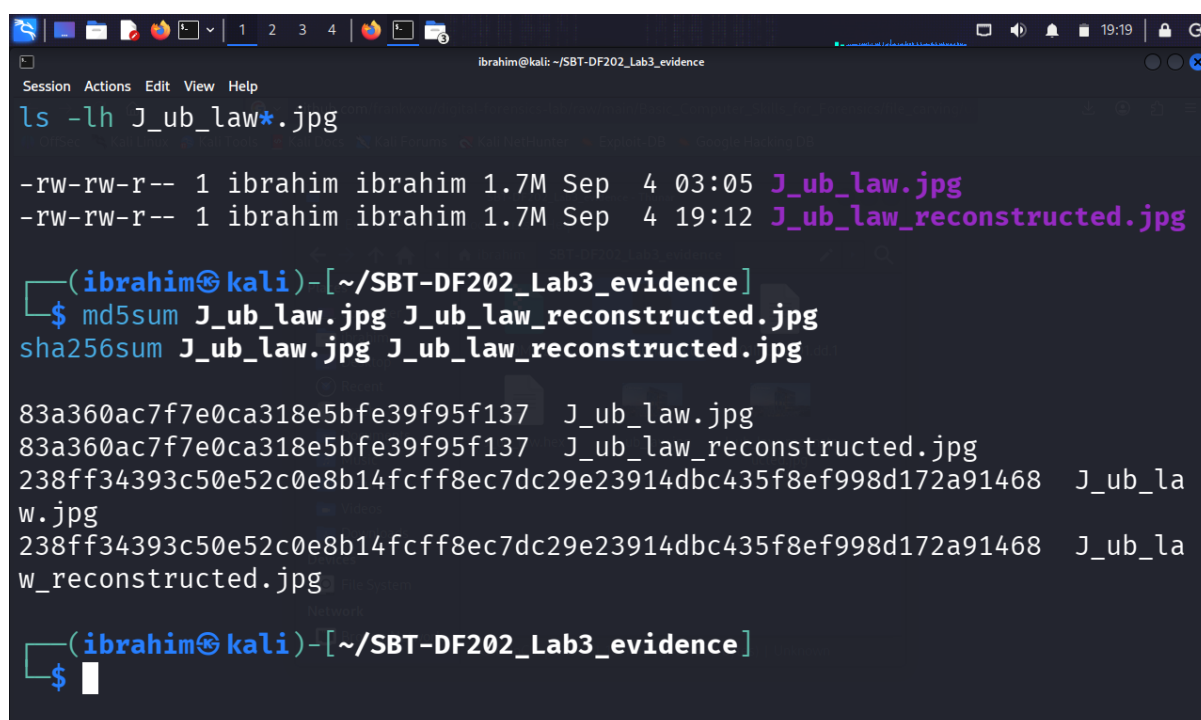
```

Figure 2.4: Reverse-hexdump reconstruction.

3.5 Verify the Reconstructed Image

```
md5sum J_ub_Law.jpg J_ub_Law_reconstructed.jpg
sha256sum J_ub_Law.jpg J_ub_Law_reconstructed.jpg
```

File	MD5 Hash	Match?
Original J_ub_law.jpg	238ff34393c50e52c0e8b14fcff8ec7dc29e23914 dbc435f8ef998d172a91468	√
Reconstructed J_ub_law.jpg	238ff34393c50e52c0e8b14fcff8ec7dc29e23914 dbc435f8ef998d172a91468	√



```

Session Actions Edit View Help
ibrahim@kali: ~/SBT-DF202_Lab3_evidence
ls -lh J_ub_law*.jpg

-rw-rw-r-- 1 ibrahim ibrahim 1.7M Sep  4 03:05 J_ub_law.jpg
-rw-rw-r-- 1 ibrahim ibrahim 1.7M Sep  4 19:12 J_ub_law_reconstructed.jpg

(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$ md5sum J_ub_law.jpg J_ub_law_reconstructed.jpg
sha256sum J_ub_law.jpg J_ub_law_reconstructed.jpg

83a360ac7f7e0ca318e5bfe39f95f137  J_ub_law.jpg
83a360ac7f7e0ca318e5bfe39f95f137  J_ub_law_reconstructed.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468  J_ub_la
w.jpg
238ff34393c50e52c0e8b14fcff8ec7dc29e23914dbc435f8ef998d172a91468  J_ub_la
w_reconstructed.jpg

(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$

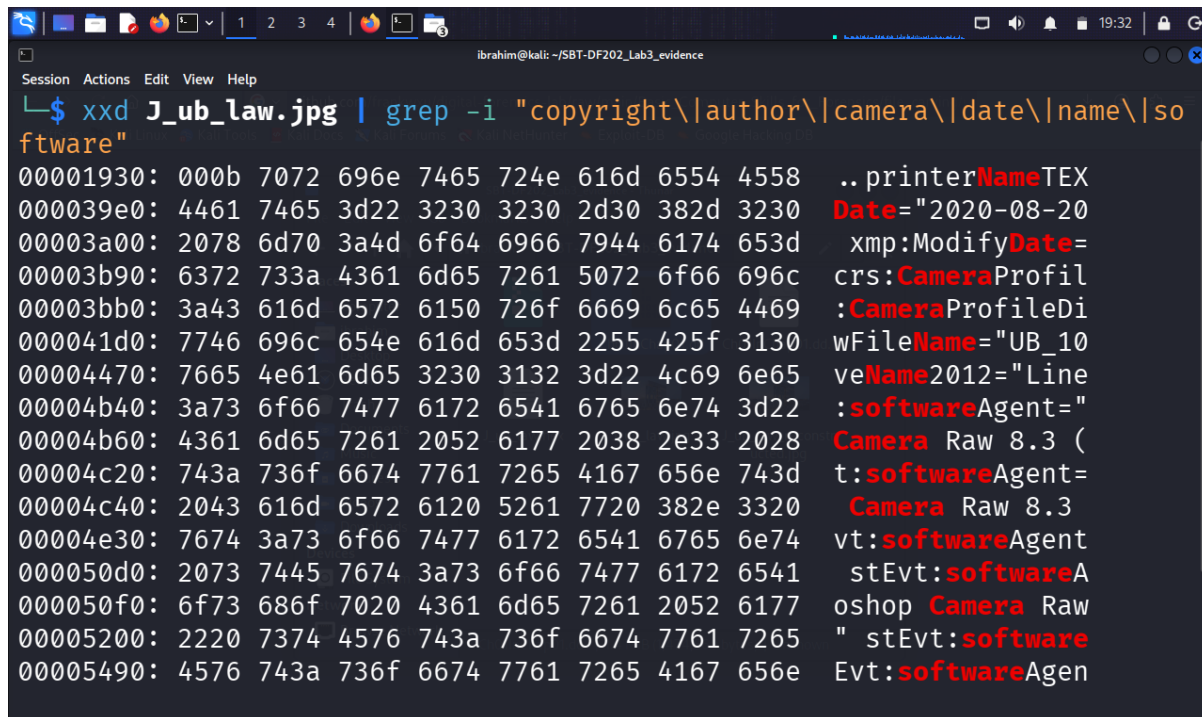
```

Figure 2.5: Hash comparison confirming successful reconstruction.

4.0 Section 4 – Part B: Examine Embedded Metadata-Like Content

4.1 Use xxd to Inspect for Strings

```
xxd J_ub_Law.jpg | grep -i "copyright\|author\|camera\|date\|name\|software"
```



```

Session Actions Edit View Help
ibrahim@kali: ~/SBT-DF202_Lab3_evidence
└─$ xxd J_ub_Law.jpg | grep -i "copyright\|author\|camera\|date\|name\|software"
00001930: 000b 7072 696e 7465 724e 616d 6554 4558  ..printerNameTEX
000039e0: 4461 7465 3d22 3230 3230 2d30 382d 3230  Date="2020-08-20
00003a00: 2078 6d70 3a4d 6f64 6966 7944 6174 653d  xmp:ModifyDate=
00003b90: 6372 733a 4361 6d65 7261 5072 6f66 696c  crs:CameraProfil
00003bb0: 3a43 616d 6572 6150 726f 6669 6c65 4469  :CameraProfileDi
000041d0: 7746 696c 654e 616d 653d 2255 425f 3130  wFileName="UB_10
00004470: 7665 4e61 6d65 3230 3132 3d22 4c69 6e65  veName2012="Line
00004b40: 3a73 6f66 7477 6172 6541 6765 6e74 3d22  :softwareAgent="
00004b60: 4361 6d65 7261 2052 6177 2038 2e33 2028  Camera Raw 8.3 (
00004c20: 743a 736f 6674 7761 7265 4167 656e 743d  t:softwareAgent=
00004c40: 2043 616d 6572 6120 5261 7720 382e 3320  Camera Raw 8.3
00004e30: 7674 3a73 6f66 7477 6172 6541 6765 6e74  vt:softwareAgent
000050d0: 2073 7445 7674 3a73 6f66 7477 6172 6541  stEvt:softwareA
000050f0: 6f73 686f 7020 4361 6d65 7261 2052 6177  oshop Camera Raw
00005200: 2220 7374 4576 743a 736f 6674 7761 7265  " stEvt:software
00005490: 4576 743a 736f 6674 7761 7265 4167 656e  Evt:softwareAgen

```

Figure 3.1: xxd metadata search results.

4.2 Use strings to Extract Metadata

```
strings J_ub_Law.jpg | grep -i "copyright\|author\|camera\|date\|name\|software\|creator\|artist"
```

```

(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$ strings J_ub_law.jpg | grep -i "copyright\|author\|camera\|date\|name\|software\|creator\|artist"
printerNameTEXT
" id="W5M0MpCehiHzreSzNTczkc9d"?> <x:xmpmeta xmlns:x="adobe:ns:meta/" x:xmptk="Adobe XMP Core 6.0-c002 79.164352, 2020/01/30-15:50:38" > <rdf:RDF xmlns:rdf="http://www.w3.org/1999/02/22-rdf-syntax-ns#" > <rdf:Description rdf:about="" xmlns:aux="http://ns.adobe.com/exif/1.0/aux/" xmlns:dc="http://purl.org/dc/elements/1.1/" xmlns:photoshop="http://ns.adobe.com/photoshop/1.0/" xmlns:xmp="http://ns.adobe.com/xap/1.0/" xmlns:xmpMM="http://ns.adobe.com/xap/1.0/mm/" xmlns:stRef="http://ns.adobe.com/xap/1.0/sType/ResourceRef#" xmlns:stEvt="http://ns.adobe.com/xap/1.0/sType/ResourceEvent#" xmlns:crs="http://ns.adobe.com/camera-raw-settings/1.0/" xmlns:lr="http://ns.adobe.com/lightroom/1.0/" aux:ApproximateFocusDistance="4294967295/1" aux:ImageNumber="115422" aux:Lens="17.0-35.0 mm f/2.8" aux:LensID="99" aux:LensInfo="170/10 350/10 28/10 28/10" aux:SerialNumber="2027750" dc:format="image/jpeg" photoshop:DateCreated="2013-10-08" photoshop:L

<?xpacket end="w"?>
Copyright 1999 Adobe Systems Incorporated

(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$

```

Figure 3.2: strings metadata extraction results.

Findings – Metadata Extracted from J_ub_law.jpg

Metadata Item	Found?	Value
Camera Model	Yes	NIKON D4
Lens	Yes	17.0-35.0 mm f/2.8
Camera Serial Number	Yes	2027750
Image Number	Yes	115422
Date/Time Created	Yes	2013-10-08T18:56:21
Date Modified	Yes	2020-08-20T10:20:05-04:00
Author/Creator	Yes	HOWARD KORN
Copyright	Yes	HOWARD KORN
Software Used	Yes	Adobe Photoshop Camera Raw 8.3, Adobe Photoshop CS6, Adobe Photoshop 21.1
Keywords	Yes	2013, campus, Merrick School of Business
Original Filename	Yes	UB_10-8-13_cam2_0117.jpg
Color Profile	Yes	Adobe RGB (1998)

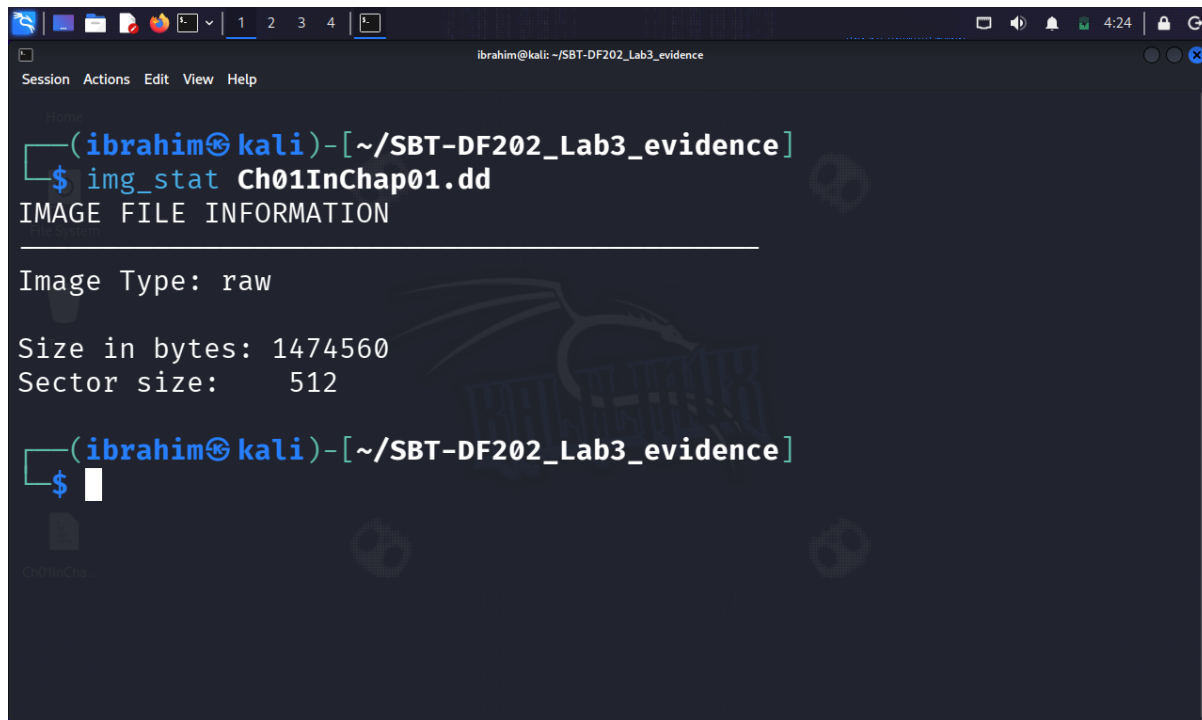
Forensic Significance

Aspect	Forensic Value
Device Identification	Camera model (NIKON D4) and serial number (2027750) link the image to a specific device.
Timeline	Date and time stamps (2013-10-08T18:56:21) establish when the image was created.
Ownership	Copyright holder (HOWARD KORN) identifies the image owner.
Software History	Multiple Adobe Photoshop edits indicate the image was modified over time.
Location Context	Keywords ("campus", "Merrick School of Business") provide location context.

5.0 Section 5 – Part C: Direct Data-Unit Examination with TSK

5.1 `img_stat` – Image Information

`img_stat Ch01InChap01.dd`

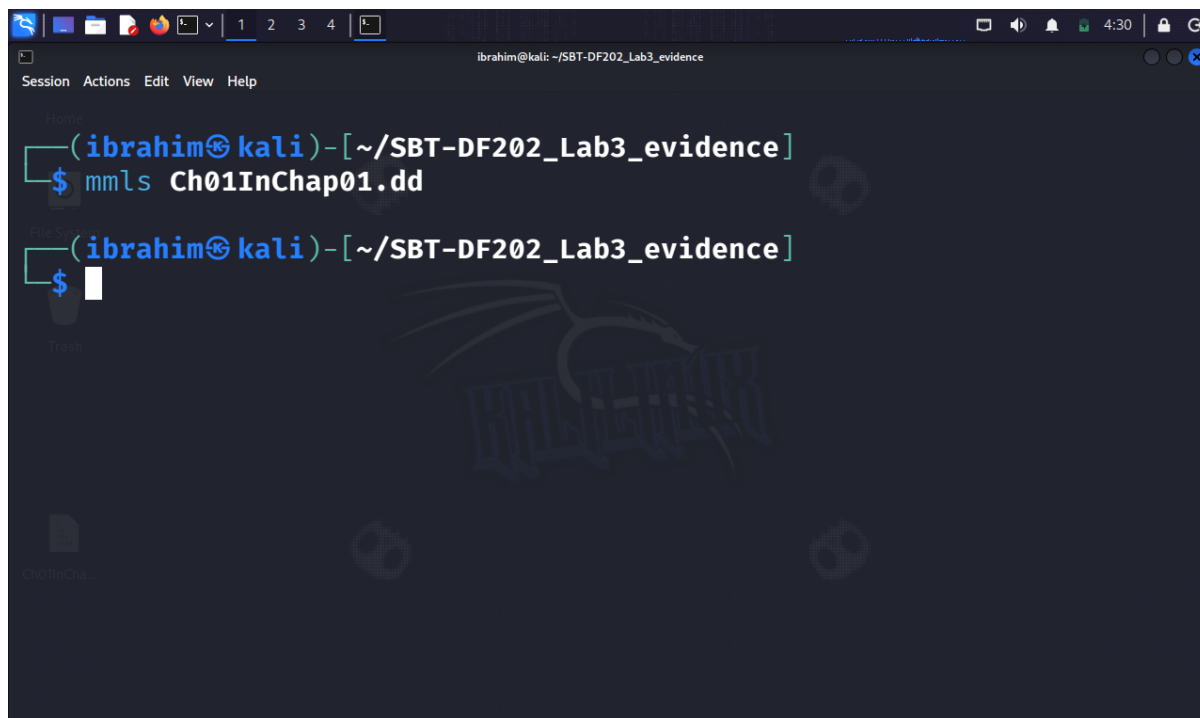


```
(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$ img_stat Ch01InChap01.dd
IMAGE FILE INFORMATION
-----
Image Type: raw
Size in bytes: 1474560
Sector size: 512
(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$
```

Figure 4.1: `img_stat` output.

5.2 `mmls` – Partition Layout

`mmls Ch01InChap01.dd`



```
ibrahim@kali: ~/SBT-DF202_Lab3_evidence
Session Actions Edit View Help

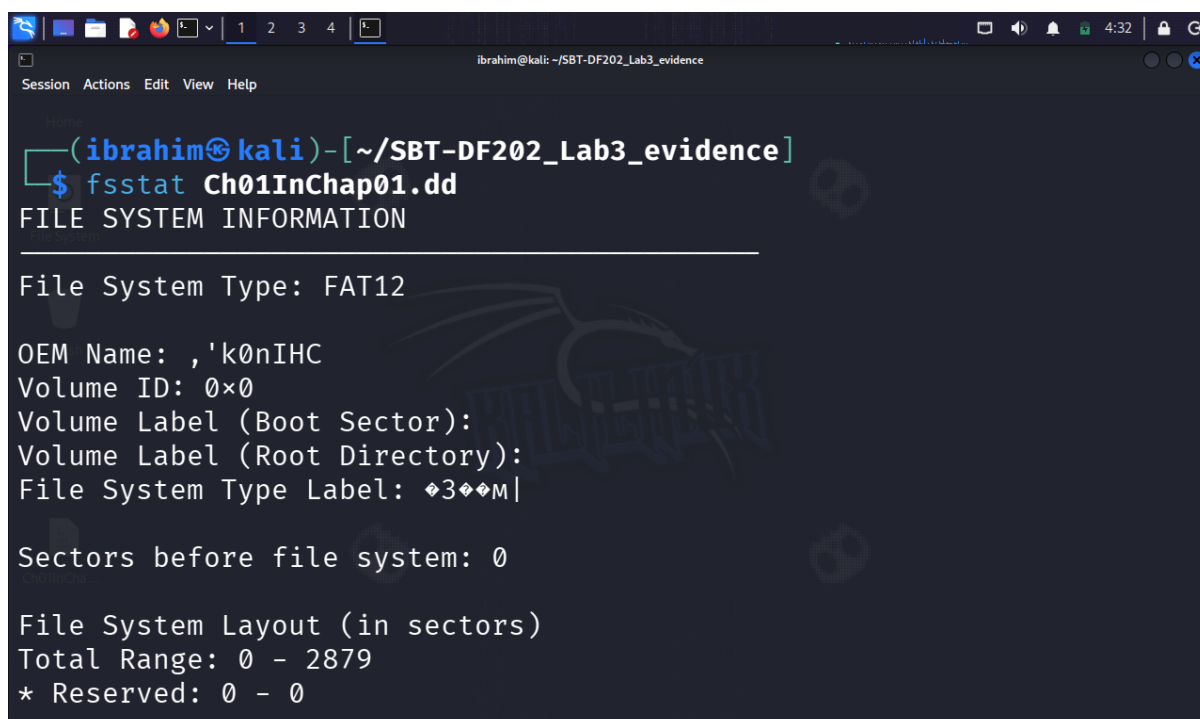
(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$ mmls Ch01InChap01.dd

(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$
```

Figure 4.2: mmls output.

5.3 fsstat – File System Information

fsstat Ch01InChap01.dd



```
ibrahim@kali: ~/SBT-DF202_Lab3_evidence
Session Actions Edit View Help

(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$ fsstat Ch01InChap01.dd
FILE SYSTEM INFORMATION
-----
File System Type: FAT12

OEM Name: , 'k0nIHC
Volume ID: 0x0
Volume Label (Boot Sector):
Volume Label (Root Directory):
File System Type Label: 3MM

Sectors before file system: 0

File System Layout (in sectors)
Total Range: 0 - 2879
* Reserved: 0 - 0
```

```

ibrahim@kali: ~/SBT-DF202_Lab3_evidence
Session Actions Edit View Help
** Root Directory: 19 - 32
** Cluster Area: 33 - 2879

METADATA INFORMATION
-----
Range: 2 - 45782
Root Directory: 2

CONTENT INFORMATION
-----
Sector Size: 512
Cluster Size: 512
Total Cluster Range: 2 - 2848

FAT CONTENTS (in sectors)
-----
33-236 (204) → EOF
285-311 (27) → EOF

```

Figure 4.3: fsstat output.

5.4 fls – File Listing

`fls Ch01InChap01.dd`

```

ibrahim@kali: ~/SBT-DF202_Lab3_evidence
Session Actions Edit View Help

(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$ fls Ch01InChap01.dd
r/r 5: Client Info.mdb
r/r * 8: Billing Letter.doc
r/r * 11: confirmation.txt
r/r 13: Income.xls
r/r * 15: letter1.txt
r/r * 17: Regrets.doc
v/v 45779: $MBR
v/v 45780: $FAT1
v/v 45781: $FAT2
V/V 45782: $OrphanFiles

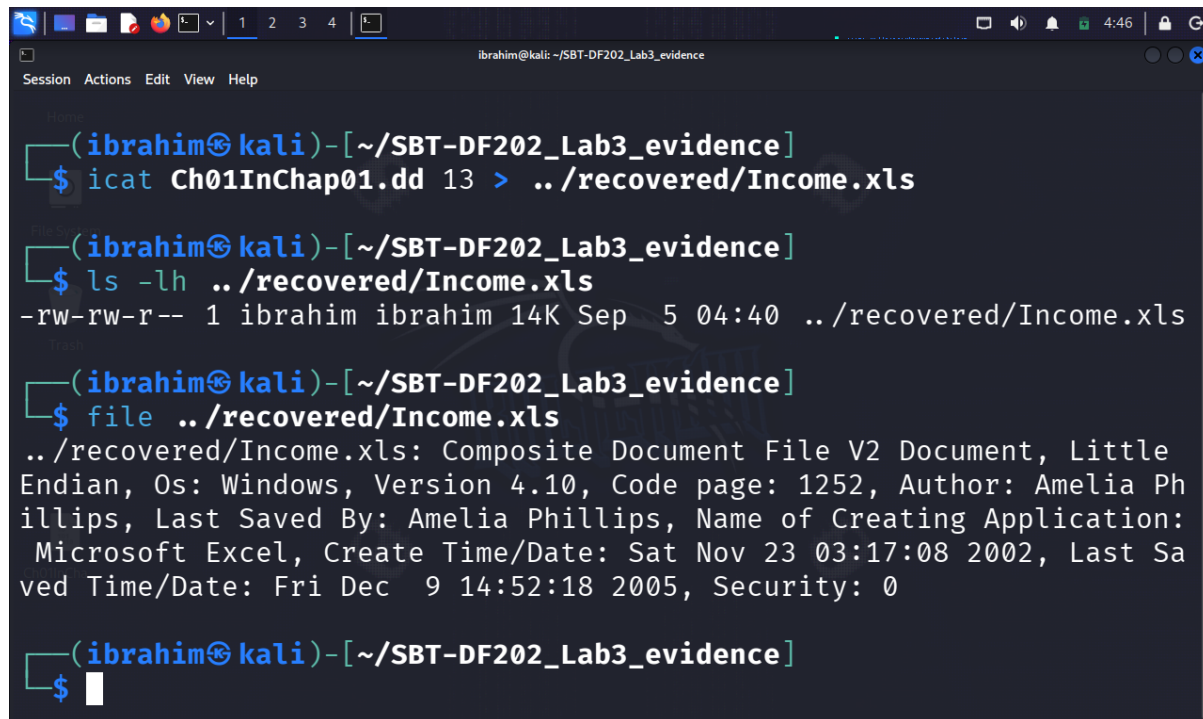
(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$ 

```

Figure 4.4: fls output showing file listing.

5.5 icat – File Recovery

```
icat Ch01InChap01.dd 13 > ../recovered/Income.xls  
ls -lh ../recovered/Income.xls  
file ../recovered/Income.xls
```

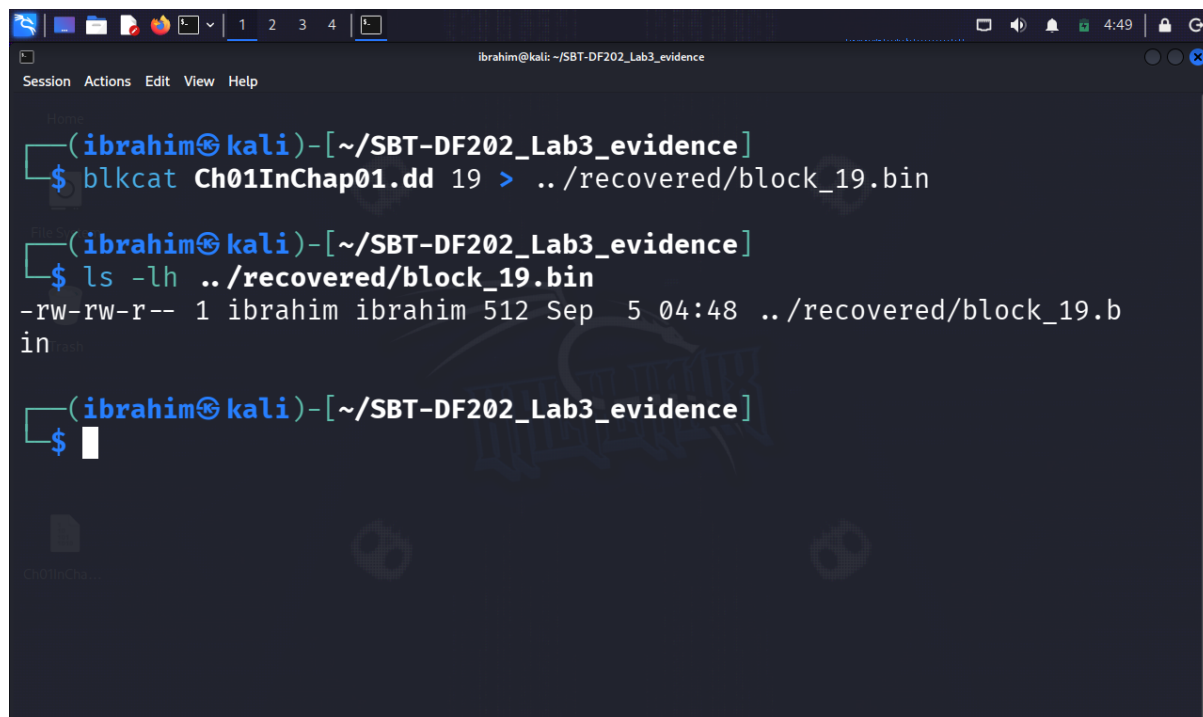


```
(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]  
$ icat Ch01InChap01.dd 13 > ../recovered/Income.xls  
  
(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]  
$ ls -lh ../recovered/Income.xls  
-rw-rw-r-- 1 ibrahim ibrahim 14K Sep  5 04:40 ../recovered/Income.xls  
  
(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]  
$ file ../recovered/Income.xls  
../recovered/Income.xls: Composite Document File V2 Document, Little  
Endian, Os: Windows, Version 4.10, Code page: 1252, Author: Amelia Ph  
illips, Last Saved By: Amelia Phillips, Name of Creating Application:  
Microsoft Excel, Create Time/Date: Sat Nov 23 03:17:08 2002, Last Sa  
ved Time/Date: Fri Dec  9 14:52:18 2005, Security: 0  
  
(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]  
$
```

Figure 4.5: icat file recovery – Income.xls successfully recovered and verified.

5.6 blkcat – Block-Level Data Extraction

```
blkcat Ch01InChap01.dd 19 > ../recovered/block_19.bin
```



```
(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]  
$ blkcat Ch01InChap01.dd 19 > ../recovered/block_19.bin  
  
(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]  
$ ls -lh ../recovered/block_19.bin  
-rw-rw-r-- 1 ibrahim ibrahim 512 Sep  5 04:48 ../recovered/block_19.b  
in  
  
(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]  
$
```

Figure 4.6: blkcat block extraction – block 19 successfully extracted.

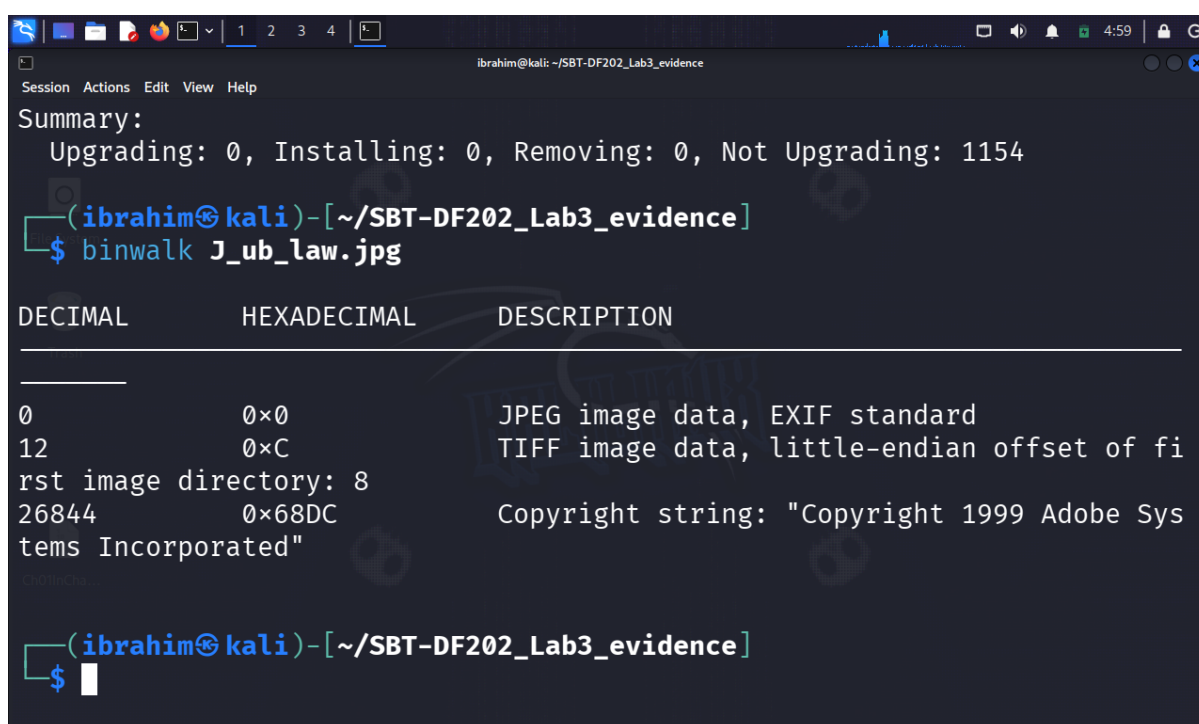
6.0 Section 6 – Part D: Embedded File Discovery with Binwalk

6.1 Install Binwalk (If Not Installed)

```
sudo apt install binwalk -y
```

6.2 Run Binwalk on the JPEG Image

```
binwalk J_ub_law.jpg
```



```

Session Actions Edit View Help
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 1154
(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$ binwalk J_ub_law.jpg

DECIMAL      HEXADECIMAL  DESCRIPTION
-----
0             0x0          JPEG image data, EXIF standard
12            0xC          TIFF image data, little-endian offset of fi
rst image directory: 8
26844         0x68DC       Copyright string: "Copyright 1999 Adobe Sys
tems Incorporated"

(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$ 

```

Figure 5.1: Binwalk output showing embedded structures.

6.3 Run Binwalk with Detailed Analysis

```
binwalk -e J_ub_law.jpg
```

```

(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$ binwalk -e J_ub_law.jpg

DECIMAL          HEXADECIMAL      DESCRIPTION
-----
WARNING: One or more files failed to extract: either no utility was found
or it's unimplemented

(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$

```

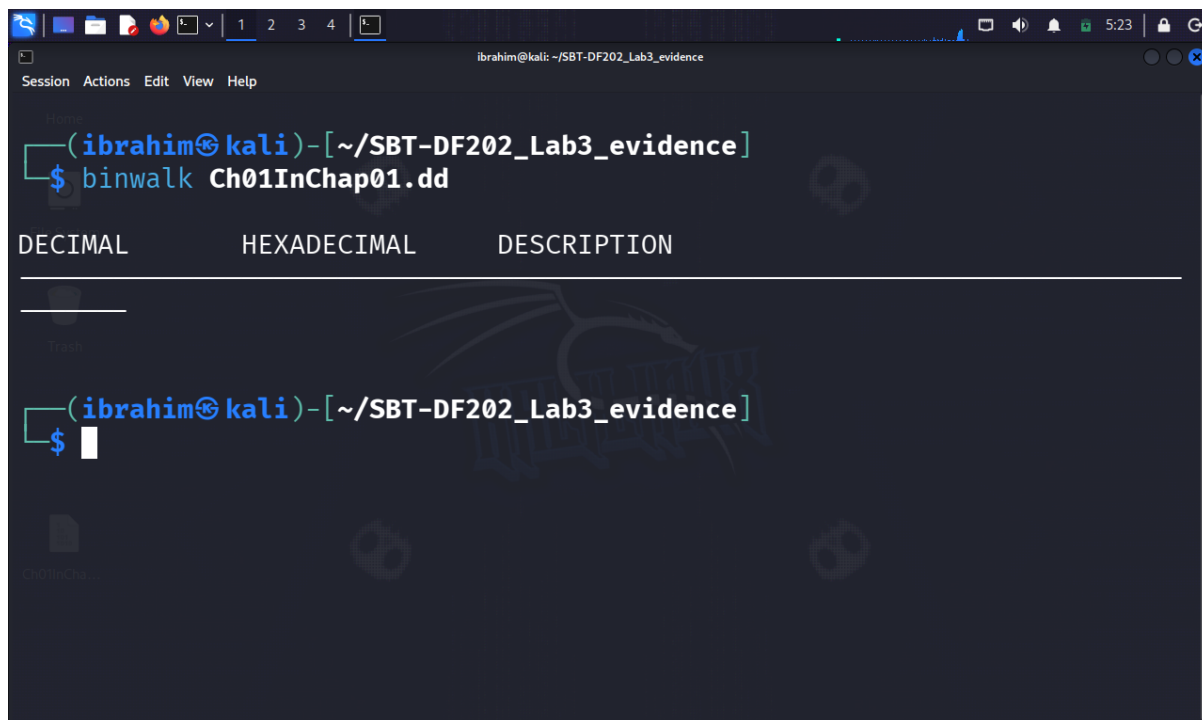
Figure 5.2: Binwalk extraction results – no embedded files found.

Interpretation:

The J_ub_law.jpg image contains no embedded file structures. The extraction attempt did not yield any recoverable files. This is a valid finding and indicates that the JPEG is a standard image without hidden embedded content.

6.4 Binwalk on the DD Image

```
binwalk Ch01InChap01.dd
```



```

(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$ binwalk Ch01InChap01.dd

DECIMAL      HEXADECIMAL    DESCRIPTION
-----
(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$

```

Figure 5.3: Binwalk output on the forensic image.

Interpretation:

The binwalk analysis of Ch01InChap01.dd did not detect any embedded file structures or recognizable file signatures. This indicates that the forensic image contains only standard file system data without hidden or nested content.

7.0 Section 7 – Part E: Prepare the USB Image

7.1 Extract 120M.7z

7z x 120M.7z
ls -lh



```
ibrahim@kali: ~/SBT-DF202_Lab3_evidence
Session Actions Edit View Help

(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$ 7z x 120M.7z
ls -lh

7-Zip 26.01 (x64) : Copyright (c) 1999-2026 Igor Pavlov : 2026-04-27
64-bit locale=en_US.UTF-8 Threads:2 OPEN_MAX:4096, ASM

Scanning the drive for archives:
1 file, 36720470 bytes (36 MiB)

Extracting archive: 120M.7z
--
Path = 120M.7z
Type = 7z
Physical Size = 36720470
Headers Size = 214
Method = LZMA2:24

Headers Size = 214
Method = LZMA2:24
Solid = +
Blocks = 1

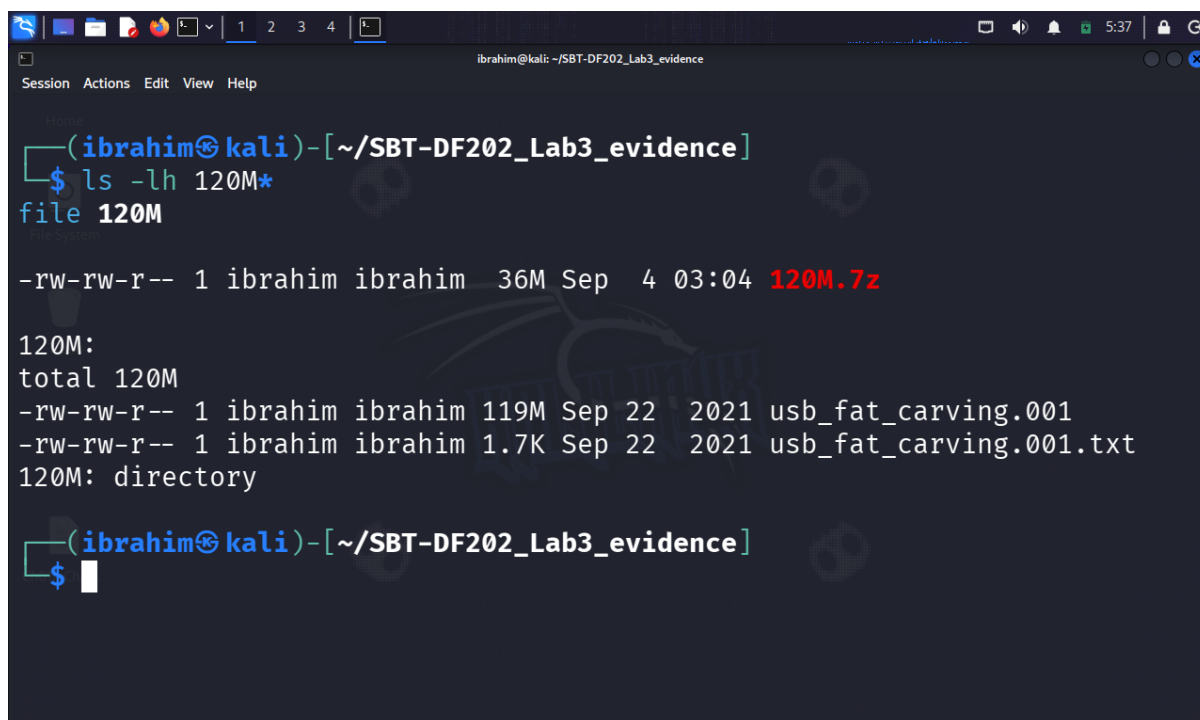
Everything is Ok

Folders: 1
Files: 2
Size:      124782229
Compressed: 36720470
total 48M
drwxrwxr-x 2 ibrahim ibrahim 4.0K Sep 22 2021 120M
-rw-rw-r-- 1 ibrahim ibrahim 36M Sep 4 03:04 120M.7z
-rwxrwx--- 1 ibrahim ibrahim 1.5M Aug 25 17:23 Ch01InChap01.dd
-rw-rw-r-- 1 ibrahim ibrahim 1.5M Sep 4 06:03 Ch01InChap01.dd.1
-rw-rw-r-- 1 ibrahim ibrahim 6.9M Sep 4 19:10 J_ub_law.hex
-rw-rw-r-- 1 ibrahim ibrahim 1.7M Sep 4 03:05 J_ub_law.jpg
```

Figure 6.1: Extraction of 120M.7z.

7.2 Identify the USB Image

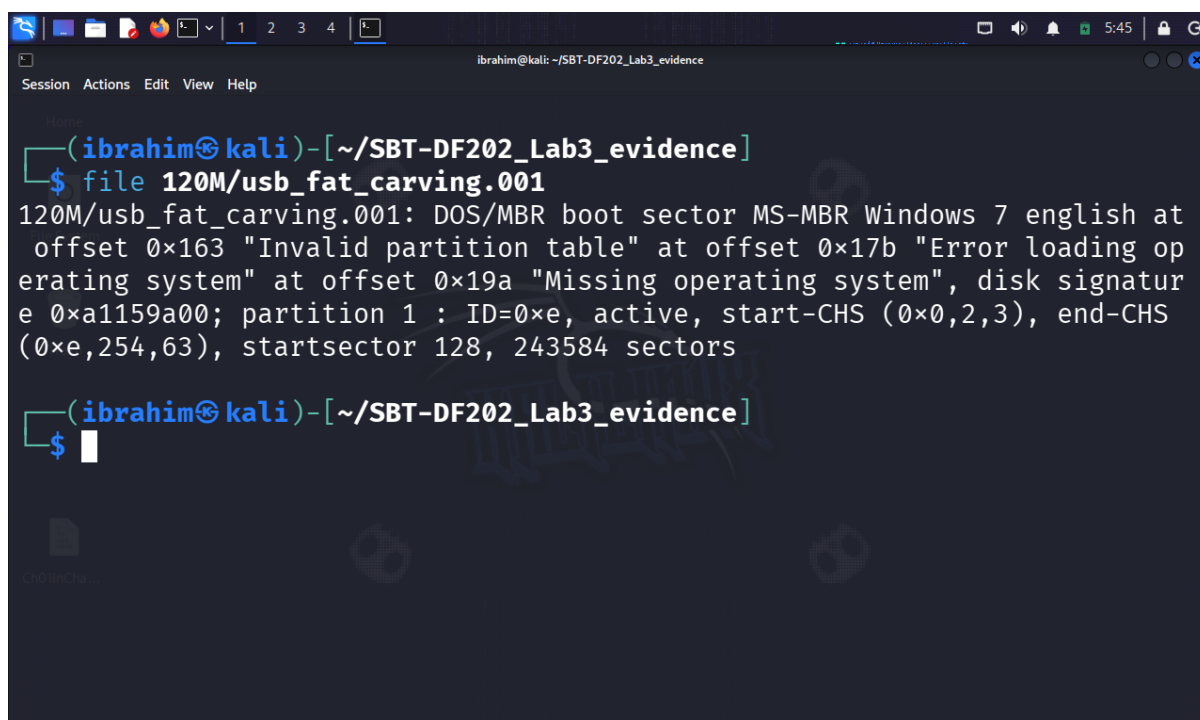
```
ls -lh 120M*
file 120M*
file 120M/usb_fat_carving.001
```



```
ibrahim@kali: ~/SBT-DF202_Lab3_evidence
$ ls -lh 120M*
-rw-rw-r-- 1 ibrahim ibrahim 36M Sep  4 03:04 120M.7z

120M:
total 120M
-rw-rw-r-- 1 ibrahim ibrahim 119M Sep 22  2021 usb_fat_carving.001
-rw-rw-r-- 1 ibrahim ibrahim 1.7K Sep 22  2021 usb_fat_carving.001.txt
120M: directory

ibrahim@kali: ~/SBT-DF202_Lab3_evidence
$
```



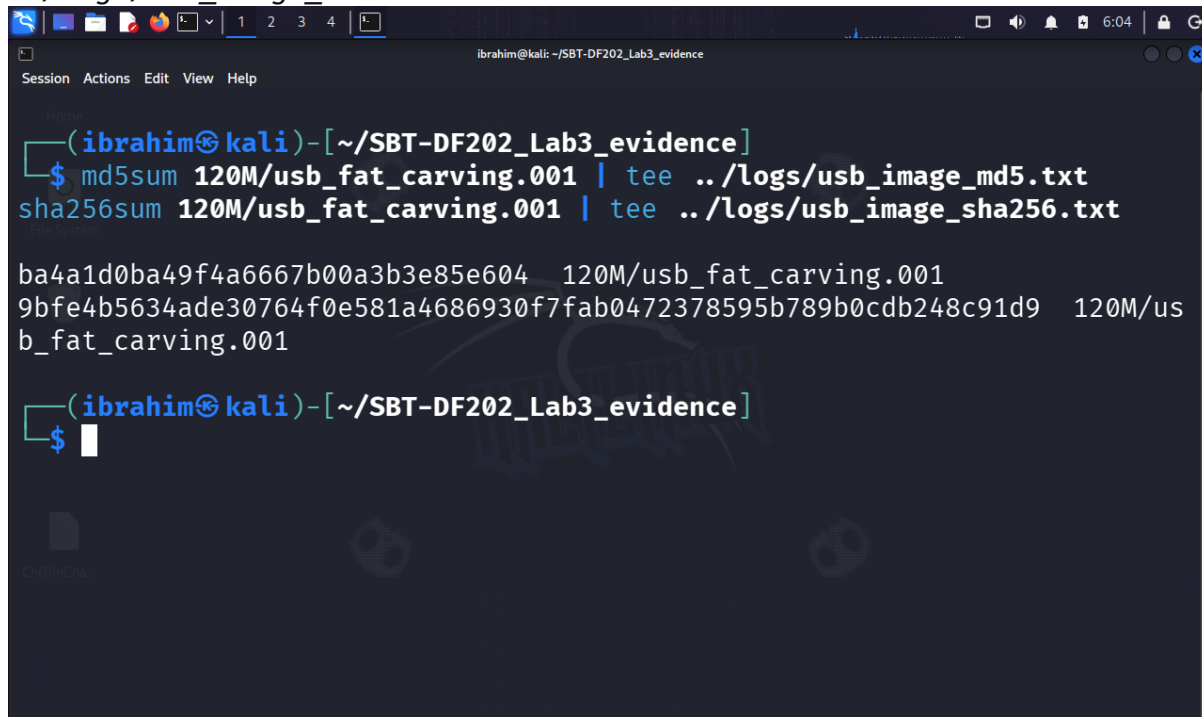
```
ibrahim@kali: ~/SBT-DF202_Lab3_evidence
$ file 120M/usb_fat_carving.001
120M/usb_fat_carving.001: DOS/MBR boot sector MS-MBR Windows 7 english at
offset 0x163 "Invalid partition table" at offset 0x17b "Error loading op
erating system" at offset 0x19a "Missing operating system", disk signatur
e 0xa1159a00; partition 1 : ID=0xe, active, start-CHS (0x0,2,3), end-CHS
(0xe,254,63), startsector 128, 243584 sectors

ibrahim@kali: ~/SBT-DF202_Lab3_evidence
$
```

Figure 6.2: USB image identification.

7.3 Calculate Hashes of the USB Image

```
md5sum 120M/usb_fat_carving.001 | tee ../logs/usb_image_md5.txt
sha256sum 120M/usb_fat_carving.001 | tee
../logs/usb_image_sha256.txt
```



```
(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$ md5sum 120M/usb_fat_carving.001 | tee ../logs/usb_image_md5.txt
sha256sum 120M/usb_fat_carving.001 | tee ../logs/usb_image_sha256.txt

ba4a1d0ba49f4a6667b00a3b3e85e604 120M/usb_fat_carving.001
9bfe4b5634ade30764f0e581a4686930f7fab0472378595b789b0cdb248c91d9 120M/us
b_fat_carving.001

(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$
```

Figure 6.3: Hashes of the USB image.

7.4 Inspect the USB Image with TSK

```
img_stat 120M/usb_fat_carving.001
mmls 120M/usb_fat_carving.001
fsstat 120M/usb_fat_carving.001
```

```

ibrahim@kali: ~/SBT-DF202_Lab3_evidence
$ img_stat 120M/usb_fat_carving.001
mmls 120M/usb_fat_carving.001
fsstat 120M/usb_fat_carving.001

IMAGE FILE INFORMATION
-----
Image Type: raw

Size in bytes: 124780544
Sector size: 512
DOS Partition Table
Offset Sector: 0
Units are in 512-byte sectors

Slot      Start      End          Length      Description
000:  Meta      00000000000  00000000000  00000000001  Primary Table (#0)
001:  _____ 00000000000  00000000127  00000000128  Unallocated
002:  000:000  00000000128  0000243711   0000243584   Win95 FAT16 (0x0e)
Possible encryption detected (High entropy (7.87))

```

Figure 6.4: TSK analysis of the USB image.

8.0 Section 8 – Part F: Carve Deleted Files with Scalpel

8.1 Install Scalpel (If Not Installed)

```
sudo apt install scalpel -y
```

8.2 Configure Scalpel for JPEG Carving

```
sudo cp /etc/scalpel/scalpel.conf /etc/scalpel/scalpel.conf.backup
```

```
sudo nano /etc/scalpel/scalpel.conf
```

Enable JPEG/JPG carving by uncommenting:

text

```
jpg y 5000000 \xff\xd8\xff\xe0 \xff\xd9
```

```

GNU nano 9.0 /etc/scalpel/scalpel.conf *
# Scalpel configuration file

# This configuration file controls the
# types and sizes of files that are carved by Scalpel.  Currently,
# Scalpel can read Foremost 0.69 configuration files, but Scalpel
# configuration files may not be backwards-compatible with Foremost.
# In particular, maximum file carve size under Foremost 0.69 is 4GB,
# while in the current version of Scalpel, it's 16EB (16 exabytes).

# For each file type, the configuration file
# describes the file's extension, whether the header and footer are
# case sensitive, the maximum file size, and the header and footer for
# the file. The footer field is optional, but header, size, case
# sensitivity, and extension are required. Any line that begins with a
# '#' is considered a comment and ignored. Thus, to skip a file type
# just put a '#' at the beginning of that line

^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^/ Go To Line

```

```

GNU nano 9.0 /etc/scalpel/scalpel.conf *
# AOL ART files
# art      y      150000  \x4a\x47\x04\x0e      \xcf\x77\xcb
# art      y      150000  \x4a\x47\x03\x0e      \xd0\xcb\x00\x00
#
# GIF and JPG files (very common)
# gif      y      5000000  \x47\x49\x46\x38\x37\x61      \x00\x3b
# gif      y      5000000  \x47\x49\x46\x38\x39\x61      \x00\x3b
# jpg      y      5242880  \xff\xd8\xff???Exif          \xff\xd9
# jpg      y      5242880  \xff\xd8\xff???JFIF          \xff\xd9
#
# PNG
# png      y      20000000  \x50\x4e\x47?      \xff\xfc\xfd\xfe
#
# BMP (used by MSWindows, use only if you have reason to think there are

```

Figure 7.1: Scalpel configuration file.

8.3 Create Output Directory

Command:

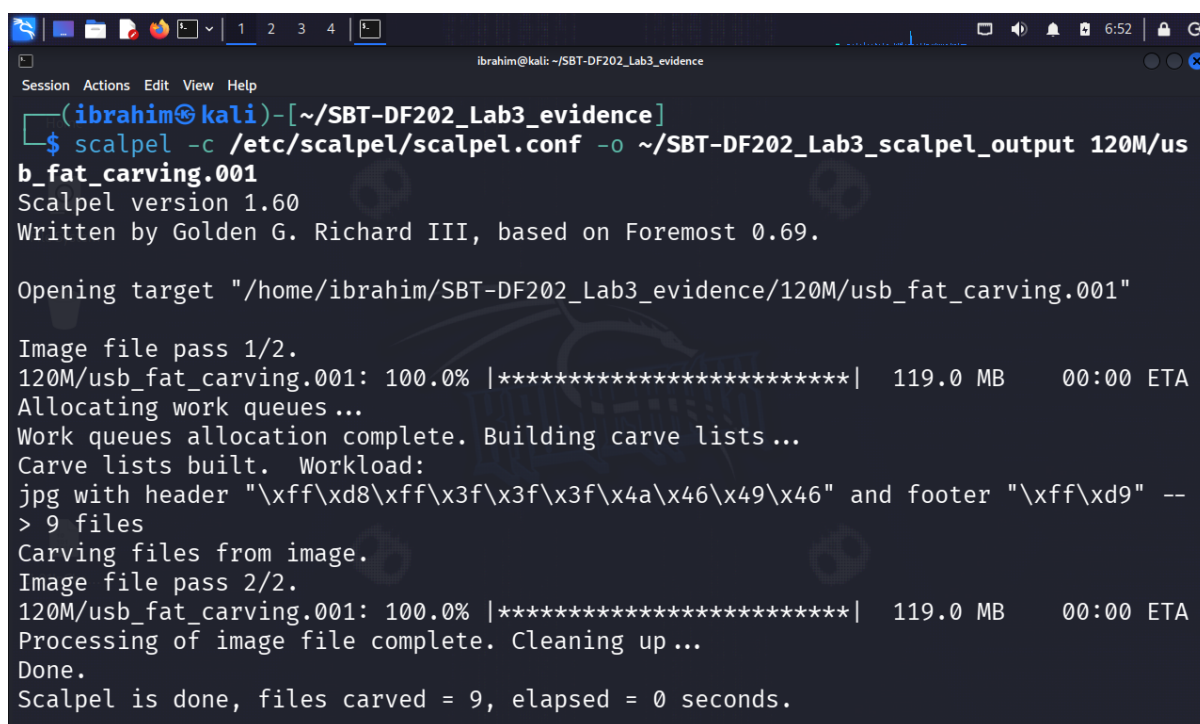
```
mkdir -p ~/SBT-DF202_Lab3_scalpel_output
```

8.4 Run Scalpel on the USB Image

IMPORTANT: Use the correct image filename `usb_fat_carving.001`, not `120M.img`.

Command:

```
scalpel -c /etc/scalpel/scalpel.conf -o ~/SBT-DF202_Lab3_scalpel_output 120M/usb_fat_carving.001
```



```
Session Actions Edit View Help
ibrahim@kali: ~/SBT-DF202_Lab3_evidence
$ scalpel -c /etc/scalpel/scalpel.conf -o ~/SBT-DF202_Lab3_scalpel_output 120M/usb_fat_carving.001
Scalpel version 1.60
Written by Golden G. Richard III, based on Foremost 0.69.

Opening target "/home/ibrahim/SBT-DF202_Lab3_evidence/120M/usb_fat_carving.001"

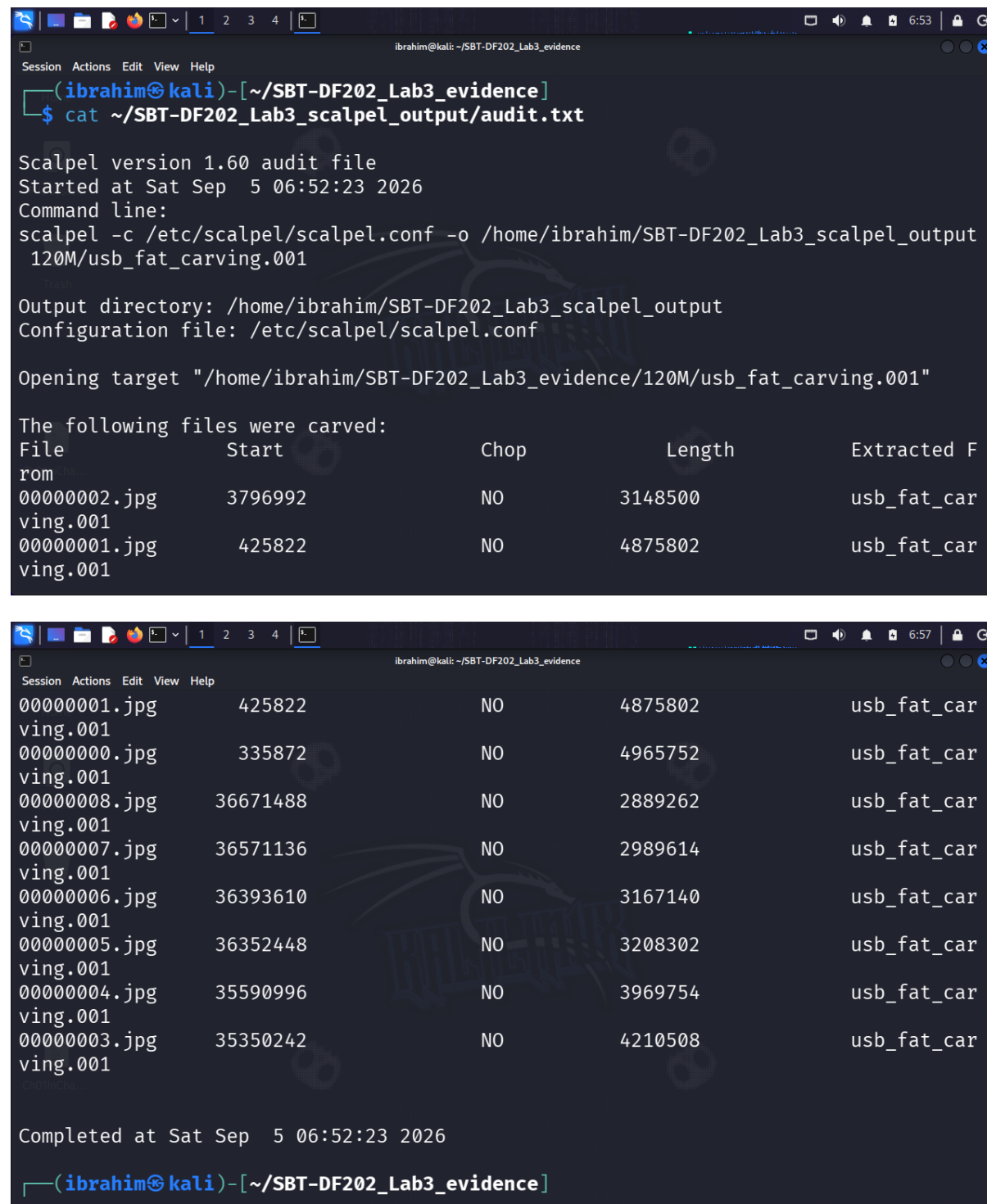
Image file pass 1/2.
120M/usb_fat_carving.001: 100.0% |*****| 119.0 MB 00:00 ETA
Allocating work queues...
Work queues allocation complete. Building carve lists...
Carve lists built. Workload:
jpg with header "\xff\xd8\xff\x3f\x3f\x3f\x4a\x46\x49\x46" and footer "\xff\xd9" --
> 9 files
Carving files from image.
Image file pass 2/2.
120M/usb_fat_carving.001: 100.0% |*****| 119.0 MB 00:00 ETA
Processing of image file complete. Cleaning up...
Done.
Scalpel is done, files carved = 9, elapsed = 0 seconds.
```

Figure 7.2: Scalpel carving execution.

8.5 Review Audit.txt

Command:

```
cat ~/SBT-DF202_Lab3_scalpel_output/audit.txt
```



```

Session Actions Edit View Help
ibrahim@kali: ~/SBT-DF202_Lab3_evidence
$ cat ~/SBT-DF202_Lab3_scalpel_output/audit.txt

Scalpel version 1.60 audit file
Started at Sat Sep 5 06:52:23 2026
Command line:
scalpel -c /etc/scalpel/scalpel.conf -o /home/ibrahim/SBT-DF202_Lab3_scalpel_output
120M/usb_fat_carving.001

Output directory: /home/ibrahim/SBT-DF202_Lab3_scalpel_output
Configuration file: /etc/scalpel/scalpel.conf

Opening target "/home/ibrahim/SBT-DF202_Lab3_evidence/120M/usb_fat_carving.001"

The following files were carved:
File                Start          Chop          Length          Extracted F
rom
00000002.jpg        3796992        NO            3148500          usb_fat_car
ving.001
00000001.jpg        425822         NO            4875802          usb_fat_car
ving.001

00000001.jpg        425822         NO            4875802          usb_fat_car
ving.001
00000000.jpg        335872         NO            4965752          usb_fat_car
ving.001
00000008.jpg        36671488       NO            2889262          usb_fat_car
ving.001
00000007.jpg        36571136       NO            2989614          usb_fat_car
ving.001
00000006.jpg        36393610       NO            3167140          usb_fat_car
ving.001
00000005.jpg        36352448       NO            3208302          usb_fat_car
ving.001
00000004.jpg        35590996       NO            3969754          usb_fat_car
ving.001
00000003.jpg        35350242       NO            4210508          usb_fat_car
ving.001

Completed at Sat Sep 5 06:52:23 2026

ibrahim@kali: ~/SBT-DF202_Lab3_evidence

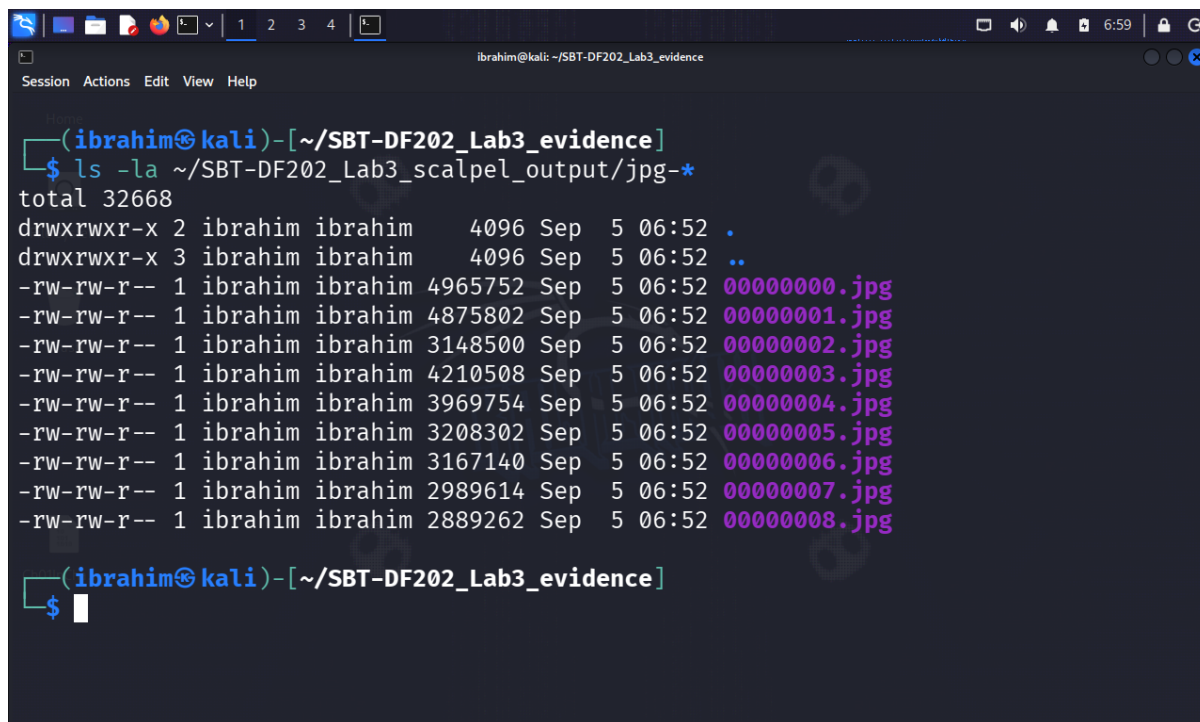
```

Figure 7.3: Scalpel audit.txt output.

8.6 Review Recovered Files

Command:

```
ls -la ~/SBT-DF202_Lab3_scalpel_output/jpg-*
```



```
(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$ ls -la ~/SBT-DF202_Lab3_scalpel_output/jpg-*
total 32668
drwxrwxr-x 2 ibrahim ibrahim 4096 Sep  5 06:52 .
drwxrwxr-x 3 ibrahim ibrahim 4096 Sep  5 06:52 ..
-rw-rw-r-- 1 ibrahim ibrahim 4965752 Sep  5 06:52 00000000.jpg
-rw-rw-r-- 1 ibrahim ibrahim 4875802 Sep  5 06:52 00000001.jpg
-rw-rw-r-- 1 ibrahim ibrahim 3148500 Sep  5 06:52 00000002.jpg
-rw-rw-r-- 1 ibrahim ibrahim 4210508 Sep  5 06:52 00000003.jpg
-rw-rw-r-- 1 ibrahim ibrahim 3969754 Sep  5 06:52 00000004.jpg
-rw-rw-r-- 1 ibrahim ibrahim 3208302 Sep  5 06:52 00000005.jpg
-rw-rw-r-- 1 ibrahim ibrahim 3167140 Sep  5 06:52 00000006.jpg
-rw-rw-r-- 1 ibrahim ibrahim 2989614 Sep  5 06:52 00000007.jpg
-rw-rw-r-- 1 ibrahim ibrahim 2889262 Sep  5 06:52 00000008.jpg

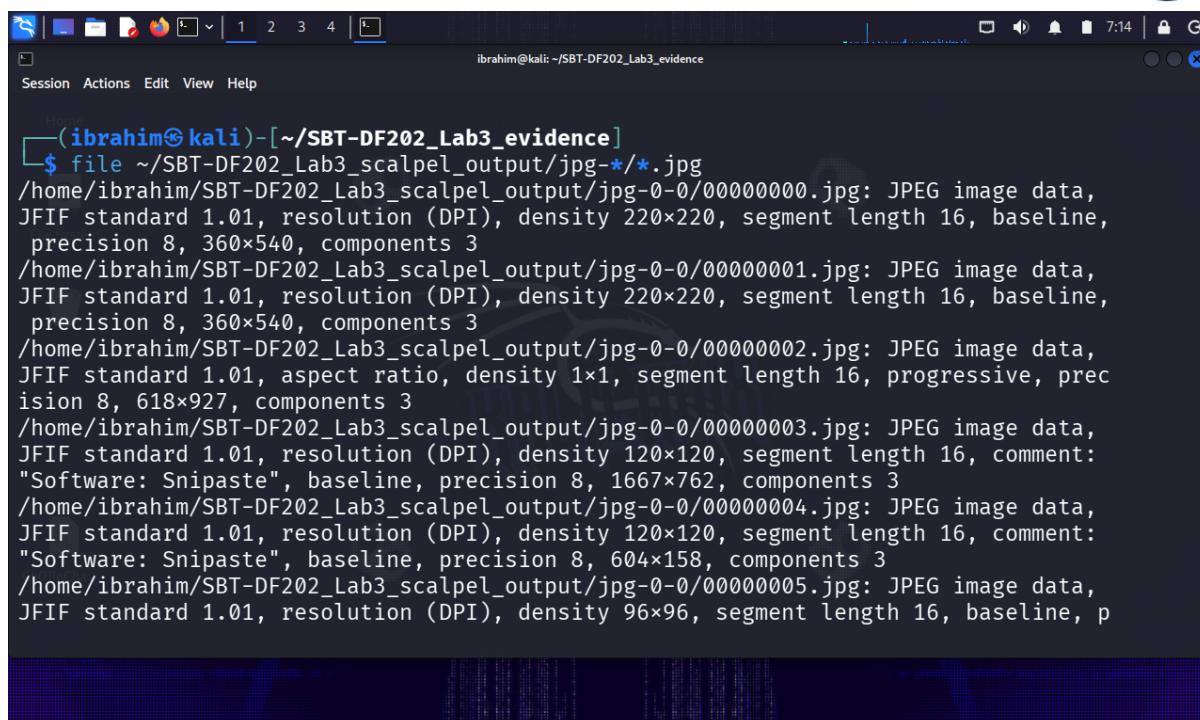
(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$
```

Figure 7.4: Recovered JPEG files.

8.7 Verify Recovered JPEG Files

Commands:

```
file ~/SBT-DF202_Lab3_scalpel_output/jpg-*/*.jpg
```



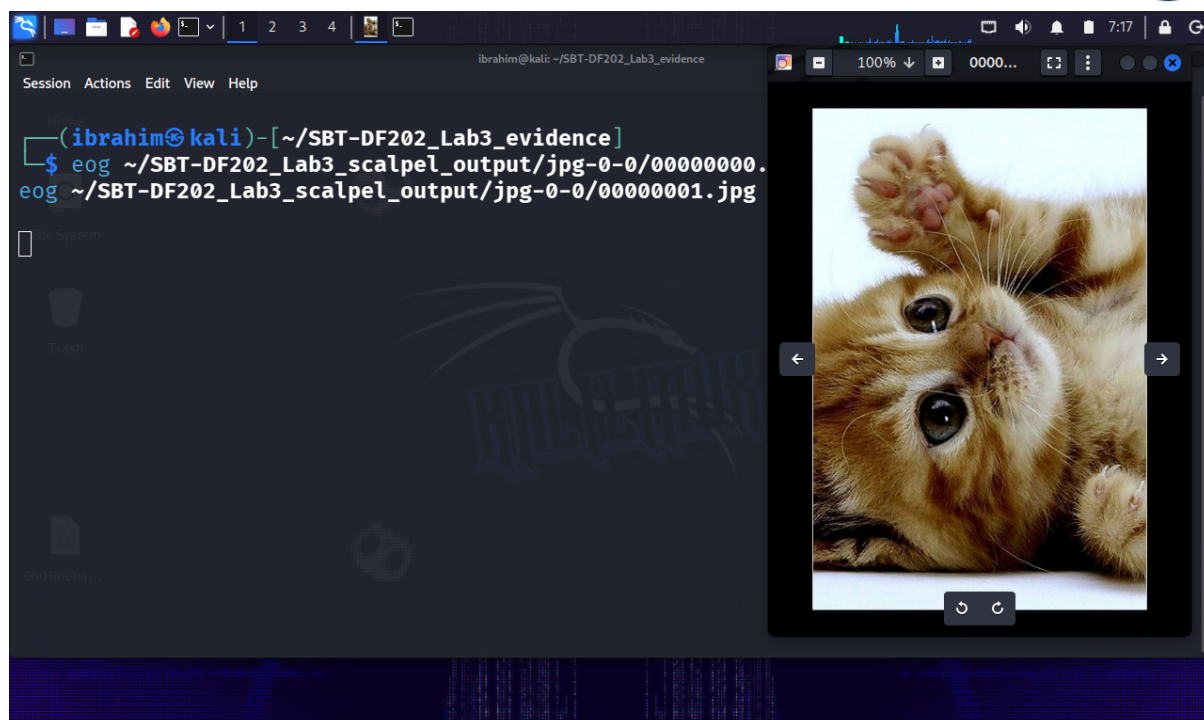
```
(ibrahim@kali)-[~/SBT-DF202_Lab3_evidence]
$ file ~/SBT-DF202_Lab3_scalpel_output/jpg-*/*.jpg
/home/ibrahim/SBT-DF202_Lab3_scalpel_output/jpg-0-0/00000000.jpg: JPEG image data,
JFIF standard 1.01, resolution (DPI), density 220x220, segment length 16, baseline,
precision 8, 360x540, components 3
/home/ibrahim/SBT-DF202_Lab3_scalpel_output/jpg-0-0/00000001.jpg: JPEG image data,
JFIF standard 1.01, resolution (DPI), density 220x220, segment length 16, baseline,
precision 8, 360x540, components 3
/home/ibrahim/SBT-DF202_Lab3_scalpel_output/jpg-0-0/00000002.jpg: JPEG image data,
JFIF standard 1.01, aspect ratio, density 1x1, segment length 16, progressive, prec
ision 8, 618x927, components 3
/home/ibrahim/SBT-DF202_Lab3_scalpel_output/jpg-0-0/00000003.jpg: JPEG image data,
JFIF standard 1.01, resolution (DPI), density 120x120, segment length 16, comment:
"Software: Snipaste", baseline, precision 8, 1667x762, components 3
/home/ibrahim/SBT-DF202_Lab3_scalpel_output/jpg-0-0/00000004.jpg: JPEG image data,
JFIF standard 1.01, resolution (DPI), density 120x120, segment length 16, comment:
"Software: Snipaste", baseline, precision 8, 604x158, components 3
/home/ibrahim/SBT-DF202_Lab3_scalpel_output/jpg-0-0/00000005.jpg: JPEG image data,
JFIF standard 1.01, resolution (DPI), density 96x96, segment length 16, baseline, p
```

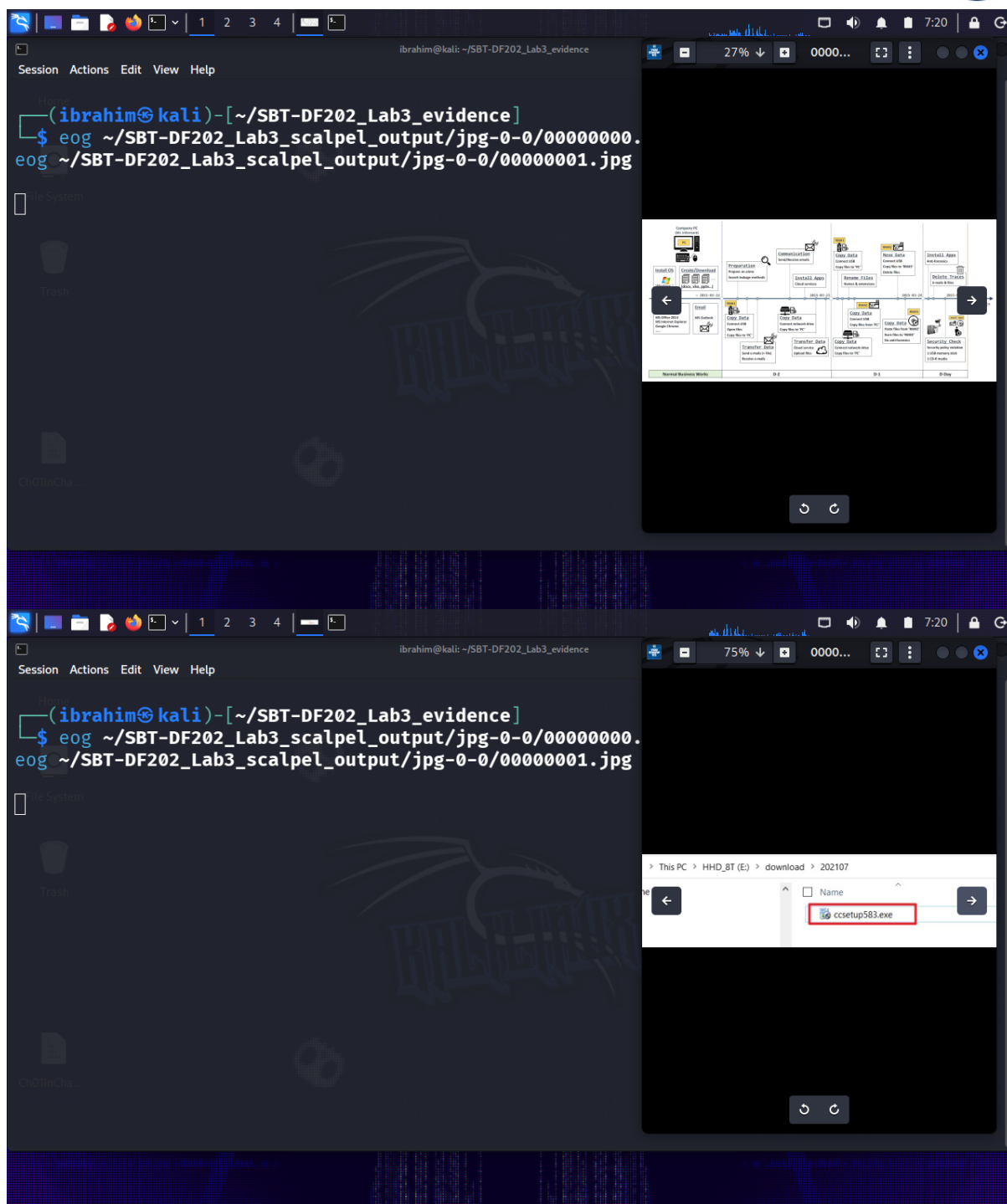
Figure 7.5: Hash verification of recovered files.

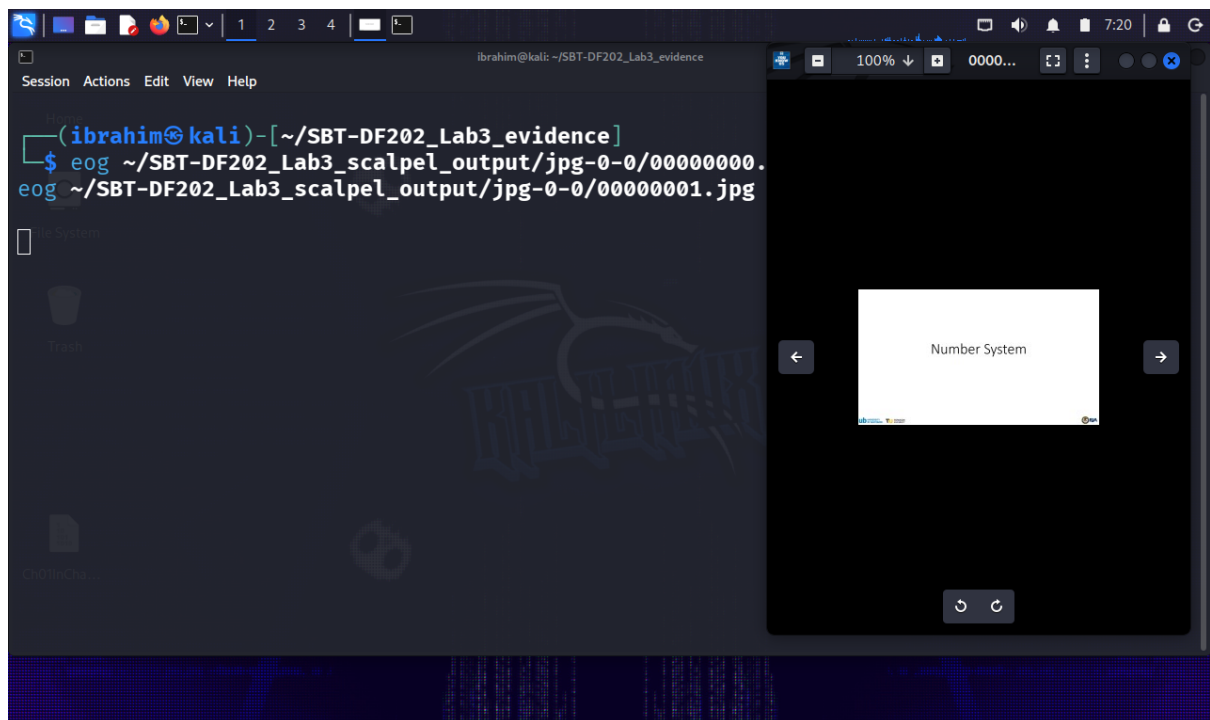
8.8 Open Recovered Images

Command:

```
eog ~/SBT-DF202_Lab3_scalpel_output/jpg-0-0/00000000.jpg
eog ~/SBT-DF202_Lab3_scalpel_output/jpg-0-0/00000001.jpg
```







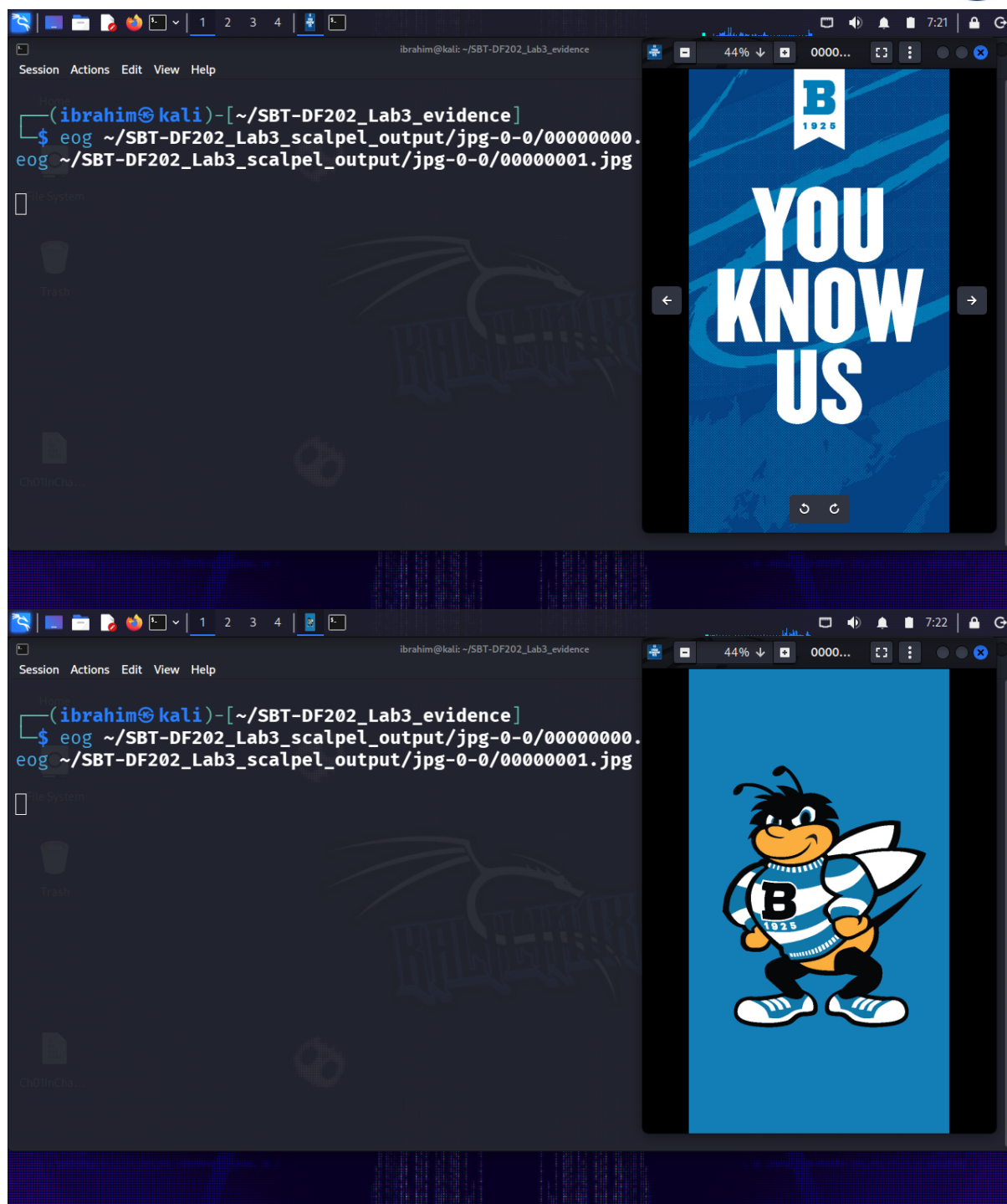


Figure 7.6: Recovered JPEG images opened successfully.



9.0 Section 9 – Conclusion

This lab provided practical experience in data carving using xxd, Binwalk, and Scalpel. I successfully:

1. Reviewed the Module 7 material before starting the lab.
2. Created a structured forensic working environment.
3. Downloaded and verified all evidence files with MD5 and SHA-256 hashes.
4. Identified JPEG file signatures (SOI: FF D8, EOI: FF D9) using xxd.
5. Created a hex dump and reconstructed the image using reverse-hexdump.
6. Verified the reconstructed image using cryptographic hashes.
7. Extracted metadata-like content using xxd and strings.
8. Used The Sleuth Kit tools to examine data units.
9. Used Binwalk to identify embedded file structures.
10. Extracted and prepared the USB image from 120M.7z.
11. Configured Scalpel and carved deleted files from the USB image.
12. Verified recovered files using cryptographic hashes.
13. Opened at least two recovered JPEG images successfully.

These skills are essential for any digital forensics professional, as they provide the foundation for recovering data when file system metadata is unavailable or unreliable.

10.0 References

1. ICDFA. (2026). SBT-DF202 – Module 7: Data Carving and File Recovery Techniques – Course Materials.
2. xxd Manual. (2026). xxd – make a hex dump or do the reverse. <https://man7.org/linux/man-pages/man1/xxd.1.html>
3. Binwalk Documentation. (2026). Binwalk – Firmware Analysis Tool. <https://github.com/ReFirmLabs/binwalk>
4. Scalpel Documentation. (2026). Scalpel – Forensic File Carving Tool. <https://github.com/digital-forensics/scalpel>
5. Sleuth Kit Documentation. (2026). The Sleuth Kit User Guide. <https://www.sleuthkit.org/sleuthkit/>
6. NIST. (2014). Guide to Integrating Forensic Techniques into Incident Response. NIST SP 800-86.

11.0 Appendices – Screenshot Reference List

Figure	Description
Figure 1.1	Lab folder structure created successfully
Figure 1.2	Evidence files downloaded
Figure 1.3	MD5 and SHA-256 hashes of evidence files
Figure 2.1	JPEG header (SOI) showing FF D8
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Figure 7.6	Recovered JPEG images opened successfully

(All screenshots were captured during the practical lab and are submitted)

Submission Statement

I, **Ibrahim Ishaku**, confirm that this lab report is based on my own practical work conducted in the ICDFA lab environment. All forensic analysis, hex inspection, data carving, and hash verification tasks are my own original work. I confirm that the original evidence files were not modified during the investigation and that all recovered evidence, screenshots, and command outputs were stored separately from the source files.

Signature: _____

Date: 4th September, 2026

End of Lab Report